Bota Andrei-Daniel

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Education

Technical University of Cluj-Napoca

Faculty of Electronics, Telecommunications and Information Technology
Major: Applied Electronics
Year of Study: 4th Year
Honors: Merit Scholarship for Academic Performance (2024–2025)

Cluj-Napoca
(2022 - present)

Solomon Haliță Theoretical High School

Specialization: Mathematics and Computer Science

Sângeorz-Băi,Bistrița-Năsăud
2018-2022

PROFESSIONAL EXPERIENCE

Internship Analog-Devices

Collaborated on the development of an integrated hardware-software system.
Applied theoretical knowledge of electronics and embedded programming to real-world industrial challenges.
Conducted technical analysis, testing, and documentation in a professional R&D environment.

July 2025 – August 2025

Service Staff

Managed customer orders and interactions in a fast-paced environment.
Developed strong communication, teamwork, and quick problem-solving skills.

Popeye Fast-Food
Bistrița-Năsăud
16/07/2019 – 30/08/2019

RELEVANT TECHNICAL PROJECTS

Hardware-Software Module with FPGA Cora Z7 and Embedded Linux Driver for AD5592R
Description: Designed and developed a complete hardware module featuring an ADC and accelerometer.
Role: Implemented the full workflow: Schematic & PCB design (KiCad), HDL implementation (Verilog) for FPGA Cora Z7, and developed an Embedded Linux SPI driver for the AD5592R.
Key Results: Delivered a functional hardware module with precise tilt response (stable output range 1.14V–2.04V).
Developed a high- level Python application for real-time data processing (mouse control, roll/pitch angles).
Tools & Technologies: KiCad, Vivado, Verilog, FPGA, Linux Kernel, SPI, GitLab CI/CD, Python.
Link: https://github.com/BotaAndreiD/analogdevicesummerschool

Embedded ECG Acquisition System — Bachelor's Thesis (ongoing)
Description: Designing a low-cost, single-lead ECG acquisition system for consumer home use, targeting cardiac monitoring accessibility at under 80 RON BOM cost.
Role: Designed the full AFE chain (INA, RLD, high-pass/low-pass/notch filters), verified with real ECG input signals.
Currently working on IEC 60601-2-25 standard compliance.
Next phases: PCB miniaturization, external ADC, dedicated MCU, and wireless PC transmission for real-time visualization.
Key Results: Functional AFE delivering clean ECG output.
Tools & Technologies: LTspice, KiCad, Embedded C, Phyton.
Link: *https://github.com/BotaAndreiD/Licenta*

Additional Personal Projects: https://github.com/BotaAndreiD/Proiecte-Personale

Additional Training

Machine Learning Course: https://github.com/BotaAndreiD/Machine-Learning-Course
Fundamentals of machine learning, data preprocessing, and model implementation (classification, regression) using Python and scikit-learn.
Analog Devices Workshops: https://github.com/BotaAndreiD/AnalogDevicesWorkshops
Introduction to analog electronics, control systems, and embedded development on Linux and bare-metal.
SMART PRACTICE Project Certificate: https://github.com/BotaAndreiD/SMART-PRACTICE-Project-Certificate
Professional training program focused on aligning academic knowledge with labor market requirements through practical internships.

Technical Skills & Languages

Programming Languages: C, C++, Java, Python, Verilog/HDL
Software & Tools: MATLAB, OrCAD, Proteus, Keil, Vivado (Xilinx), LTspice, KiCad, GitLab CI/CD, GitHub
Hardware Platforms: FPGA (Cora Z7, Artix-7), ADALM2000, Analog Discovery, Scopy
Languages: English (B2), Romanian (Native), German (A1)